

Tutorial Sheet-3

Indian Institute of Technology Jodhpur

MAL1020 (Mathematics-II)

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1. Let $T : V \rightarrow V$ be a linear transformation. So prove that $T^2 = 0$ iff $\text{Range}(T) \subseteq \text{Null}(T)$.
2. Explain whether there exist a linear transformation $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ such that

$$T(0, 1, 1, 1) = (2, 0)$$

$$T(1, 2, 1, 1) = (1, 2)$$

$$T(1, 1, 1, 2) = (3, 1)$$

$$T(2, 1, 0, 1) = (2, 3)$$

3. Consider the map $T : \mathbb{R}^4 \rightarrow \mathbb{R}^2$ by $T(x, y, z, w) = (x + y, z + w)$. Check if T is a linear transformation. Find the matrix associated with T w.r.t. standard basis of $\mathbb{R}^4$ and $\mathbb{R}^2$.
4. Examine whether the following maps $T : V \rightarrow W$ are linear transformations.
 - (a) $V = W = \mathbb{C}^2(\mathbb{C})$ and $T[z_1, z_2]^T = [\bar{z}_1, \bar{z}_2]^T$ for all $[z_1, z_2]^T \in \mathbb{C}^2$
 - (b) $V = W = M_n(\mathbb{R})$. For a fix $B \in M_n(\mathbb{R})$, define $T(A) = AB - BA$ for all $A \in V$.
5. Find bases for the range space and the null space of the linear transformation $T : M_2(\mathbb{R}) \rightarrow M_2(\mathbb{R})$ defined by $T(A) = AB - BA$ for all $A \in M_2(\mathbb{R})$, where $B = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$. Hence find the rank and the nullity of T .
6. Consider the linear transformation $T : \mathbb{R}^3 \rightarrow \mathbb{R}^2$, define by

$$T[x, y, z]^T = [2x + 3y - z, 4x - y + 2z]^T \text{ for all } [x, y, z]^T \in \mathbb{R}^3$$

Find $[T]_{C,B}$, where $B = \{[1, 1, 0]^T, [1, 2, 3]^T, [1, 3, 5]^T\}$ and $C = \{[1, 2]^T, [2, 3]^T\}$.

7. Examine whether $\mathbf{T}$ is a linear mapping. If $\mathbf{T}$ is linear, find $\text{Ker } \mathbf{T}$ and $\text{Im } \mathbf{T}$.
 - (i) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ defined by $T(x, y) = (x + y, x - y)$, $(x, y) \in \mathbb{R}^2$;
 - (ii) $T : \mathbb{R}^2 \rightarrow \mathbb{R}^3$ defined by $T(x, y) = (x + 2y, 2x + y, x + y)$, $(x, y) \in \mathbb{R}^2$;
 - (iii) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (yz, zx, xy)$, $(x, y, z) \in \mathbb{R}^3$;
 - (iv) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ defined by $T(x, y, z) = (x + 2y + 3z, 3x + 2y + z, x + y + z)$, $(x, y, z) \in \mathbb{R}^3$;
 - (v) $T : \mathbb{R}^3 \rightarrow \mathbb{R}^4$ defined by $T(x, y, z) = (-x + y + z, x - y + z, x + y - z, x + y + z)$, $(x, y, z) \in \mathbb{R}^3$;
 - (vi) $T : \mathbb{R}^3 \rightarrow \mathbb{R}$ defined by $T(x, y, z) = x + y + z$, $(x, y, z) \in \mathbb{R}^3$;
 - (vii) $T : \mathbb{R}_{2 \times 2} \rightarrow \mathbb{R}_{2 \times 2}$ defined by $T(A) = \frac{1}{2}(A + A^T)$, $A \in \mathbb{R}_{2 \times 2}$.
8. Determine the linear mapping $\mathbf{T} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that maps the basis vectors $(1, 0, 0)$, $(0, 1, 0)$, $(0, 0, 1)$ of $\mathbb{R}^3$ to the vectors $(0, 1, 0)$, $(0, 0, 1)$, $(0, 1, 0)$ respectively. Find $\text{Ker } \mathbf{T}$ and $\text{Im } \mathbf{T}$. Verify that $\dim \text{Ker } \mathbf{T} + \dim \text{Im } \mathbf{T} = 3$.

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9. Determine the linear mapping $\mathbf{T} : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ that maps the basis vectors $(0, 1, 1)$, $(1, 0, 1)$, $(1, 1, 0)$ of $\mathbb{R}^3$ to the vectors $(2, 1, 1)$, $(1, 2, 1)$, $(1, 1, 2)$ respectively. Find $\text{Ker } \mathbf{T}$ and $\text{Im } \mathbf{T}$. Verify that $\dim \text{Ker } \mathbf{T} + \dim \text{Im } \mathbf{T} = 3$.
10. Find a linear operator $\mathbf{T}$ on $\mathbb{R}^3$ such that $\text{Ker } \mathbf{T}$ is the subspace $U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\}$ of $\mathbb{R}^3$.
11. A linear mapping $\mathbf{T} : \mathbb{R}^3 \rightarrow \mathbb{R}^2$ is defined by $T(x_1, x_2, x_3) = (3x_1 - 2x_2 + x_3, x_1 - 3x_2 - 2x_3)$, $(x_1, x_2, x_3) \in \mathbb{R}^3$. Find the matrix of $\mathbf{T}$ relative to the ordered bases
- (i) $\{(1, 0, 0), (0, 1, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$;
 - (ii) $\{(0, 1, 0), (1, 0, 0), (0, 0, 1)\}$ of $\mathbb{R}^3$ and $\{(0, 1), (1, 0)\}$ of $\mathbb{R}^2$;
 - (iii) $\{(0, 1, 1), (1, 0, 1), (1, 1, 0)\}$ of $\mathbb{R}^3$ and $\{(1, 0), (0, 1)\}$ of $\mathbb{R}^2$.
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